Probabilità e Statistica per l'Informatica

sez. E - P / Lezione 16

Alessandro Toigo

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Densità condizionate (caso discreto)

Definizione

(i) $\vec{X} : \Omega \to S_{\vec{X}}$: vettore aleatorio discreto

(ii) $Y : \Omega \to S_Y$: variabile aleatoria discreta

(iii) $\ell \in S_Y$ t.c. $p_Y(\ell) \neq 0$

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La *densità condizionata* di $\vec{X}$ dato $Y = \ell$ è la funzione

$$p_{\vec{X}\,|\,Y}(\,\cdot\,|\,\ell) : S_{\vec{X}} \to \mathbb{R}\,, \qquad p_{\vec{X}\,|\,Y}(\vec{k}\,|\,\ell) = \frac{p_{(\vec{X},Y)}(\vec{k},\ell)}{p_Y(\ell)}$$

dove $(\vec{X}, Y) = (X_1, \ldots, X_n, Y)$

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PROPRIETÀ:

- $p_{\vec{X}\,|\,Y}(\,\cdot\,|\,\ell)$ = densità di $\vec{X}$ rispetto a $\mathbb{P}' = \mathbb{P}(\,\cdot\,|\,\text{“}Y = \ell\text{”})$

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Densità condizionate (caso discreto)

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$$= \mathbb{P}\big(\vec{X} = \vec{k}\,|\,\text{“}Y = \ell\text{”}\big)$$

Densità condizionate (caso discreto)

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$$= \mathbb{P}\big(\vec{X} = \vec{k}\,|\,\text{“}Y = \ell\text{”}\big) = \mathbb{P}'(\vec{X} = \vec{k})$$

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$$= \mathbb{P}\big(\vec{X}=\vec{k}\,|\,Y=\ell\big) = \mathbb{P}'(\vec{X}=\vec{k})$$

D'ora in poi $\mathbb{P}(\,\cdot\,|\,\text{“}Y=\ell\text{”}) = \mathbb{P}(\,\cdot\,|\,Y=\ell)$

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PROPRIETÀ:

- $p_{\vec{X}\,|\,Y}(\,\cdot\,|\,\ell)$ = densità di $\vec{X}$ rispetto a $\mathbb{P}' = \mathbb{P}(\,\cdot\,|\ Y = \ell\ )$
- $p_{\vec{X}\,|\,Y}(\,\cdot\,|\,\ell)$ è positiva e normalizzata

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- $p_{\vec{X}\,|\,Y}(\,\cdot\,|\,\ell)$ è positiva e normalizzata
- se $Y = c$ è costante, allora $p_{\vec{X}\,|\,Y}(\,\cdot\,|\,c) = p_{\vec{X}}$

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  “$Y = c$” $= \Omega \quad \Rightarrow \quad \mathbb{P}(\,\cdot\,|\,Y = c) = \mathbb{P}$

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- $p_{\vec{X}}(\vec{k}) = \sum_{\ell \in S_Y} p_{\vec{X}\,|\,Y}(\vec{k}\,|\,\ell)\, p_Y(\ell)$ formula delle probabilità totali

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È un 'se e solo se' !

Densità condizionate (caso discreto)

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- X e Y indipendenti $\Leftrightarrow$ $p_{X\,|\,Y}(\,\cdot\,|\,\ell) = p_X$ per ogni ℓ:

  X, Y indip. $\Leftrightarrow$ $p_X(k)\, p_Y(\ell) = p_{(X,Y)}(k,\ell)$ $\forall k,\ell$

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- X e Y indipendenti $\Leftrightarrow$ $p_{X\,|\,Y}(\,\cdot\,|\,\ell) = p_X$ per ogni ℓ:

  X, Y indip. $\Leftrightarrow$ $p_X(\,\cdot\,)\, p_Y(\ell) = p_{(X,Y)}(k,\ell) = p_{X\,|\,Y}(\,\cdot\,|\,\ell)\, p_Y(\ell)$ $\forall k,\ell$

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- X e Y indipendenti $\Leftrightarrow$ $p_{X\,|\,Y}(\,\cdot\,|\,\ell) = p_X$ per ogni ℓ:

  X, Y indip. $\Leftrightarrow$ $p_X(\,\cdot\,)\,p_Y(\ell) = p_{(X,Y)}(k,\ell) = p_{X\,|\,Y}(\,\cdot\,|\,\ell)\,p_Y(\ell)$ $\forall k,\ell$

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PROPRIETÀ:

- $p_{(\vec{X},Y)\,|\,Y}(\vec{k},h\,|\,\ell) = p_{\vec{X}\,|\,Y}(\vec{k}\,|\,\ell)\,\delta_{h,\ell}$ dove

$$\delta_{h,\ell} = \begin{cases} 1 & \text{se } h = \ell \\ 0 & \text{se } h \neq \ell \end{cases} \qquad (\textit{delta di Kronecker})$$

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PROPRIETÀ:

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$$p_{(\vec{X},Y)\,|\,Y}(\vec{k},h\,|\,\ell) = \frac{p_{(\vec{X},Y,Y)}(\vec{k},h,\ell)}{p_Y(\ell)}$$

Densità condizionate (caso discreto)

Definizione

$$p_{\vec{X}\,|\,Y}(\,\cdot\,|\,\ell) : S_{\vec{X}} \to \mathbb{R}\,, \qquad p_{\vec{X}\,|\,Y}(\vec{k}\,|\,\ell) = \frac{p_{(\vec{X},Y)}(\vec{k},\ell)}{p_Y(\ell)}$$

PROPRIETÀ:

- $p_{(\vec{X},Y)\,|\,Y}(\vec{k},h\,|\,\ell) = p_{\vec{X}\,|\,Y}(\vec{k}\,|\,\ell)\,\delta_{h,\ell}$:

$$p_{(\vec{X},Y)\,|\,Y}(\vec{k},h\,|\,\ell) = \frac{p_{(\vec{X},Y,Y)}(\vec{k},h,\ell)}{p_Y(\ell)} = \frac{\mathbb{P}\big(\vec{X}=\vec{k},\,Y=h,\,Y=\ell\big)}{p_Y(\ell)}$$

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$$= \frac{\mathbb{P}\big(\vec{X}=\vec{k},\ Y=\ell\big)\,\delta_{h,\ell}}{p_Y(\ell)}$$

perché

$$\text{“}Y=h\text{”} \wedge \text{“}Y=\ell\text{”} \;=\; \begin{cases} \text{“}Y=\ell\text{”} & \text{se } h=\ell \\ \emptyset & \text{se } h\neq\ell \end{cases}$$

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Valore atteso condizionato (caso discreto)

Definizione

(i) X, Y : variabili aleatorie discrete

(ii) $\ell \in S_Y$ t.c. $p_Y(\ell) \neq 0$

Valore atteso condizionato (caso discreto)

Definizione

(i) X, Y : variabili aleatorie discrete

(ii) $\ell \in S_Y$ t.c. $p_Y(\ell) \neq 0$

Il *valore atteso condizionato* di X dato $Y = \ell$ è il numero reale

$$\mathbb{E}[X \mid Y = \ell] = \sum_{k \in S_X} k \, p_{X \mid Y}(k \mid \ell)$$

Valore atteso condizionato (caso discreto)

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$$\mathbb{E}[X \mid Y = \ell] = \sum_{k \in S_X} k \, p_{X \mid Y}(k \mid \ell)$$

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- $\mathbb{E}[\,\cdot \mid Y = \ell]$ ha le proprietà del valore atteso (lineare, positivo . . .)

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- $\mathbb{E}\big[g(\vec{X}) \mid Y = \ell\big] = \sum_{\vec{k} \in S_{\vec{X}}} g(\vec{k}) \, p_{\vec{X} \mid Y}(\vec{k} \mid \ell)$

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Non vale il viceversa !

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- $\mathbb{E}[X] = \sum_{\ell \in S_Y} \mathbb{E}[X \mid Y = \ell]\; p_Y(\ell)$
- X e Y indipendenti $\Rightarrow$ $\mathbb{E}[X \mid Y = \ell] = \mathbb{E}[X]$ per ogni ℓ:

  X, Y indip. $\Rightarrow$ $p_{X \mid Y}(\,\cdot \mid \ell) = p_X$

Valore atteso condizionato (caso discreto)

Definizione

$$\mathbb{E}[X \mid Y = \ell] = \sum_{k \in S_X} k \, p_{X \mid Y}(k \mid \ell)$$

PROPRIETÀ:

- $\mathbb{E}\big[g(\vec{X}, Y) \mid Y = \ell\big] = \mathbb{E}\big[g(\vec{X}, \ell) \mid Y = \ell\big]$
- $\mathbb{E}\big[g_1(\vec{X})\, g_2(Y) \mid Y = \ell\big] = g_2(\ell)\, \mathbb{E}\big[g_1(\vec{X}) \mid Y = \ell\big]$
- $\mathbb{E}[X] = \sum_{\ell \in S_Y} \mathbb{E}[X \mid Y = \ell]\; p_Y(\ell)$
- X e Y indipendenti $\Rightarrow$ $\mathbb{E}[X \mid Y = \ell] = \mathbb{E}[X]$ per ogni ℓ:

  X, Y indip. $\Rightarrow$ $p_{X \mid Y}(\,\cdot \mid \ell) = p_X$ $\Rightarrow$ $\mathbb{E}[X \mid Y = \ell] = \mathbb{E}[X]$

Valore atteso condizionato (caso discreto)

Definizione

$$\mathbb{E}[X \mid Y = \ell] = \sum_{k \in S_X} k \, p_{X \mid Y}(k \mid \ell)$$

PROPRIETÀ:

- $\mathbb{E}\big[g(\vec{X}, Y) \mid Y = \ell\big] = \mathbb{E}\big[g(\vec{X}, \ell) \mid Y = \ell\big]$
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- $\mathbb{E}[X] = \sum_{\ell \in S_Y} \mathbb{E}[X \mid Y = \ell]\; p_Y(\ell)$
- X e Y indipendenti $\Rightarrow$ $\mathbb{E}[X \mid Y = \ell] = \mathbb{E}[X]$ per ogni ℓ

Valore atteso condizionato (caso discreto)

Definizione

(i) X, Y : variabili aleatorie discrete

(ii) $\ell \in S_Y$ t.c. $p_Y(\ell) \neq 0$

Il valore atteso condizionato di X dato $Y = \ell$ è il numero reale

$$\mathbb{E}[X \mid Y = \ell] = \sum_{k \in S_X} k \, p_{X \mid Y}(k \mid \ell)$$

Valore atteso condizionato (caso discreto)

Definizione

(i) X, Y : variabili aleatorie discrete

(ii) $\ell \in S_Y$ t.c. $p_Y(\ell) \neq 0$

Il valore atteso condizionato di X dato $Y = \ell$ è la variabile aleatoria

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Valore atteso condizionato (caso discreto)

Definizione

$$\mathbb{E}[X \mid Y] = \sum_{k \in S_X} k \, p_{X \mid Y}(k \mid Y)$$

Valore atteso condizionato (caso discreto)

Definizione

$$\mathbb{E}[X \mid Y] = \sum_{k \in S_X} k \, p_{X \mid Y}(k \mid Y)$$

PROPRIETÀ:

- $\mathbb{E}[X \mid Y]$ è una variabile aleatoria

Valore atteso condizionato (caso discreto)

Definizione

$$\mathbb{E}[X \mid Y] = \sum_{k \in S_X} k \, p_{X \mid Y}(k \mid Y)$$

PROPRIETÀ:

- $\mathbb{E}[X \mid Y]$ è una variabile aleatoria
- $\mathbb{E}[X \mid Y] = e(Y)$ con $e(\ell) = \mathbb{E}[X \mid Y = \ell]$

Valore atteso condizionato (caso discreto)

Definizione

$$\mathbb{E}[X \mid Y] = \sum_{k \in S_X} k \, p_{X \mid Y}(k \mid Y)$$

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Valore atteso condizionato (caso discreto)

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$$e(\ell) = \mathbb{E}[X \mid Y = \ell] = \sum_{k \in S_X} k \, p_{X \mid Y}(k \mid \ell)$$

Valore atteso condizionato (caso discreto)

Definizione

$$\mathbb{E}[X \mid Y] = \sum_{k \in S_X} k \, p_{X \mid Y}(k \mid Y)$$

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$$e(\ell) = \mathbb{E}[X \mid Y = \ell] = \sum_{k \in S_X} k \, p_{X \mid Y}(k \mid \ell)$$

$$\Rightarrow \quad e(Y) = \sum_{k \in S_X} k \, p_{X \mid Y}(k \mid Y)$$

Valore atteso condizionato (caso discreto)

Definizione

$$\mathbb{E}[X \mid Y] = \sum_{k \in S_X} k \, p_{X \mid Y}(k \mid Y)$$

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- $\mathbb{E}[X \mid Y]$ è una variabile aleatoria
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$$e(\ell) = \mathbb{E}[X \mid Y = \ell] = \sum_{k \in S_X} k \, p_{X \mid Y}(k \mid \ell)$$

$$\Rightarrow \quad e(Y) = \sum_{k \in S_X} k \, p_{X \mid Y}(k \mid Y) = \mathbb{E}[X \mid Y]$$

Valore atteso condizionato (caso discreto)

Definizione

$$\mathbb{E}[X \mid Y] = \sum_{k \in S_X} k \, p_{X \mid Y}(k \mid Y)$$

PROPRIETÀ:

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Valore atteso condizionato (caso discreto)

Definizione

$$\mathbb{E}[X \mid Y] = \sum_{k \in S_X} k \, p_{X \mid Y}(k \mid Y)$$

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- $\mathbb{E}[X \mid Y] = \sum_{\ell \in S_Y} \mathbb{E}[X \mid Y = \ell] \cdot \mathbb{1}_{\text{“}Y=\ell\text{”}}$:

$$\mathbb{E}[X \mid Y] = e(Y)$$

Valore atteso condizionato (caso discreto)

Definizione

$$\mathbb{E}[X \mid Y] = \sum_{k \in S_X} k \, p_{X \mid Y}(k \mid Y)$$

PROPRIETÀ:

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$$\mathbb{E}[X \mid Y] = e(Y) = 1 \cdot e(Y)$$

Valore atteso condizionato (caso discreto)

Definizione

$$\mathbb{E}[X \mid Y] = \sum_{k \in S_X} k \, p_{X \mid Y}(k \mid Y)$$

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$$\mathbb{E}[X \mid Y] = e(Y) = 1 \cdot e(Y) = \sum_{\ell \in S_Y} \mathbb{1}_{\text{“}Y=\ell\text{”}} \cdot e(Y)$$

Valore atteso condizionato (caso discreto)

Definizione

$$\mathbb{E}[X \mid Y] = \sum_{k \in S_X} k \, p_{X \mid Y}(k \mid Y)$$

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Definizione

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$$\mathbb{E}[X \mid Y] = e(Y) = 1 \cdot e(Y) = \sum_{\ell \in S_Y} \mathbb{1}_{\text{“}Y=\ell\text{”}} \cdot e(Y)$$

$$= \sum_{\ell \in S_Y} \mathbb{1}_{\text{“}Y=\ell\text{”}} \cdot e(\ell) = \sum_{\ell \in S_Y} \mathbb{1}_{\text{“}Y=\ell\text{”}} \cdot \mathbb{E}[X \mid Y = \ell]$$

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Definizione

$$\mathbb{E}[X \mid Y] = \sum_{k \in S_X} k \, p_{X \mid Y}(k \mid Y)$$

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- $\mathbb{E}[a_1 X_1 + a_2 X_2 \mid Y] = a_1 \mathbb{E}[X_1 \mid Y] + a_2 \mathbb{E}[X_2 \mid Y]$ (v.a. somma)

Valore atteso condizionato (caso discreto)

Definizione

$$\mathbb{E}[X \mid Y] = \sum_{k \in S_X} k \, p_{X \mid Y}(k \mid Y)$$

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- $\mathbb{E}[a_1 X_1 + a_2 X_2 \mid Y] = a_1 \mathbb{E}[X_1 \mid Y] + a_2 \mathbb{E}[X_2 \mid Y]$
- $X \geq 0 \quad \Rightarrow \quad \mathbb{E}[X \mid Y] \geq 0$ (v.a. positiva)

Valore atteso condizionato (caso discreto)

Definizione

$$\mathbb{E}[X \mid Y] = \sum_{k \in S_X} k \, p_{X \mid Y}(k \mid Y)$$

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- $\mathbb{E}[a_1 X_1 + a_2 X_2 \mid Y] = a_1 \mathbb{E}[X_1 \mid Y] + a_2 \mathbb{E}[X_2 \mid Y]$
- $X \geq 0 \quad \Rightarrow \quad \mathbb{E}[X \mid Y] \geq 0$
- $\mathbb{E}[c \mid Y] = c$ (v.a. costante)

Valore atteso condizionato (caso discreto)

Definizione

$$\mathbb{E}[X \mid Y] = \sum_{k \in S_X} k \, p_{X \mid Y}(k \mid Y)$$

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- $\mathbb{E}[c \mid Y] = c$

Valore atteso condizionato (caso discreto)

Definizione

$$\mathbb{E}[X \mid Y] = \sum_{k \in S_X} k \, p_{X \mid Y}(k \mid Y)$$

PROPRIETÀ:

- $\mathbb{E}[\mathbb{E}[X \mid Y]] = \mathbb{E}[X]$

Valore atteso condizionato (caso discreto)

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PROPRIETÀ:

- $\mathbb{E}[\mathbb{E}[X \mid Y]] = \mathbb{E}[X]$:

$$\mathbb{E}[\mathbb{E}[X \mid Y]] = \mathbb{E}\Big[\sum_{\ell \in S_Y} \mathbb{E}[X \mid Y = \ell] \cdot \mathbb{1}_{\text{“}Y=\ell\text{”}}\Big]$$

Valore atteso condizionato (caso discreto)

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$$\mathbb{E}[X \mid Y] = \sum_{k \in S_X} k \, p_{X \mid Y}(k \mid Y)$$

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$$= \sum_{\ell \in S_Y} \mathbb{E}[X \mid Y = \ell] \; \mathbb{E}\Big[\mathbb{1}_{\text{“}Y=\ell\text{”}}\Big]$$

Valore atteso condizionato (caso discreto)

Definizione

$$\mathbb{E}[X \mid Y] = \sum_{k \in S_X} k \, p_{X \mid Y}(k \mid Y)$$

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$$= \sum_{\ell \in S_Y} \mathbb{E}[X \mid Y = \ell] \, \mathbb{E}\big[\mathbb{1}_{\text{“}Y=\ell\text{”}}\big] = \sum_{\ell \in S_Y} \mathbb{E}[X \mid Y = \ell] \, \mathbb{P}(Y = \ell)$$

Valore atteso condizionato (caso discreto)

Definizione

$$\mathbb{E}[X \mid Y] = \sum_{k \in S_X} k \, p_{X \mid Y}(k \mid Y)$$

PROPRIETÀ:

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$$= \sum_{\ell \in S_Y} \mathbb{E}[X \mid Y = \ell] \, \mathbb{E}\big[\mathbb{1}_{\text{“}Y=\ell\text{”}}\big] = \sum_{\ell \in S_Y} \mathbb{E}[X \mid Y = \ell] \, \mathbb{P}(Y = \ell)$$

$$= \sum_{\ell \in S_Y} \mathbb{E}[X \mid Y = \ell] \, p_Y(\ell)$$

Valore atteso condizionato (caso discreto)

Definizione

$$\mathbb{E}[X \mid Y] = \sum_{k \in S_X} k \, p_{X \mid Y}(k \mid Y)$$

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$$= \sum_{\ell \in S_Y} \mathbb{E}[X \mid Y = \ell] \, p_Y(\ell) = \mathbb{E}[X]$$

Valore atteso condizionato (caso discreto)

Definizione

$$\mathbb{E}[X \mid Y] = \sum_{k \in S_X} k \, p_{X \mid Y}(k \mid Y)$$

PROPRIETÀ:

- $\mathbb{E}[\mathbb{E}[X \mid Y]] = \mathbb{E}[X]$:

$$\mathbb{E}[\mathbb{E}[X \mid Y]] = \mathbb{E}\Big[\sum_{\ell \in S_Y} \mathbb{E}[X \mid Y = \ell] \cdot \mathbb{1}_{\text{“}Y=\ell\text{”}}\Big]$$

$$= \sum_{\ell \in S_Y} \mathbb{E}[X \mid Y = \ell] \, \mathbb{E}\big[\mathbb{1}_{\text{“}Y=\ell\text{”}}\big] = \sum_{\ell \in S_Y} \mathbb{E}[X \mid Y = \ell] \, \mathbb{P}(Y = \ell)$$

$$= \sum_{\ell \in S_Y} \mathbb{E}[X \mid Y = \ell] \, p_Y(\ell) = \mathbb{E}[X]$$

Valore atteso condizionato (caso discreto)

Definizione

$$\mathbb{E}[X \mid Y] = \sum_{k \in S_X} k \, p_{X \mid Y}(k \mid Y)$$

PROPRIETÀ:

- $\mathbb{E}[\mathbb{E}[X \mid Y]] = \mathbb{E}[X]$
- $\mathbb{E}[g(Y) X \mid Y] = g(Y) \mathbb{E}[X \mid Y]$

Valore atteso condizionato (caso discreto)

Definizione

$$\mathbb{E}[X \mid Y] = \sum_{k \in S_X} k \, p_{X \mid Y}(k \mid Y)$$

PROPRIETÀ:

- $\mathbb{E}[\mathbb{E}[X \mid Y]] = \mathbb{E}[X]$
- $\mathbb{E}[g(Y) X \mid Y] = g(Y) \mathbb{E}[X \mid Y]$:

$$\mathbb{E}[g(Y) X \mid Y] = \sum_{\ell \in S_Y} \mathbb{E}[g(Y) X \mid Y = \ell] \cdot \mathbb{1}_{\text{“}Y=\ell\text{”}}$$

Valore atteso condizionato (caso discreto)

Definizione

$$\mathbb{E}[X \mid Y] = \sum_{k \in S_X} k \, p_{X \mid Y}(k \mid Y)$$

PROPRIETÀ:

- $\mathbb{E}[\mathbb{E}[X \mid Y]] = \mathbb{E}[X]$
- $\mathbb{E}[g(Y)\,X \mid Y] = g(Y)\,\mathbb{E}[X \mid Y]$:

$$\mathbb{E}[g(Y)\,X \mid Y] = \sum_{\ell \in S_Y} \mathbb{E}[g(Y)\,X \mid Y = \ell] \cdot \mathbb{1}_{\text{“}Y=\ell\text{”}}$$

$$= \sum_{\ell \in S_Y} g(\ell)\ \mathbb{E}[X \mid Y = \ell] \cdot \mathbb{1}_{\text{“}Y=\ell\text{”}}$$

Valore atteso condizionato (caso discreto)

Definizione

$$\mathbb{E}[X \mid Y] = \sum_{k \in S_X} k \, p_{X \mid Y}(k \mid Y)$$

PROPRIETÀ:

- $\mathbb{E}[\mathbb{E}[X \mid Y]] = \mathbb{E}[X]$
- $\mathbb{E}[g(Y)\,X \mid Y] = g(Y)\,\mathbb{E}[X \mid Y]$:

$$\mathbb{E}[g(Y)\,X \mid Y] = \sum_{\ell \in S_Y} \mathbb{E}[g(Y)\,X \mid Y = \ell] \cdot \mathbb{1}_{\text{“}Y=\ell\text{”}}$$

$$= \sum_{\ell \in S_Y} g(\ell) \underbrace{\mathbb{E}[X \mid Y = \ell]}_{e(\ell)} \cdot \mathbb{1}_{\text{“}Y=\ell\text{”}} = \sum_{\ell \in S_Y} g(Y)\, e(Y)\, \mathbb{1}_{\text{“}Y=\ell\text{”}}$$

Valore atteso condizionato (caso discreto)

Definizione

$$\mathbb{E}[X \mid Y] = \sum_{k \in S_X} k \, p_{X \mid Y}(k \mid Y)$$

PROPRIETÀ:

- $\mathbb{E}[\mathbb{E}[X \mid Y]] = \mathbb{E}[X]$
- $\mathbb{E}[g(Y) X \mid Y] = g(Y) \mathbb{E}[X \mid Y]$:

$$\mathbb{E}[g(Y) X \mid Y] = \sum_{\ell \in S_Y} \mathbb{E}[g(Y) X \mid Y = \ell] \cdot \mathbb{1}_{\text{“}Y=\ell\text{”}}$$

$$= \sum_{\ell \in S_Y} g(\ell) \underbrace{\mathbb{E}[X \mid Y = \ell]}_{e(\ell)} \cdot \mathbb{1}_{\text{“}Y=\ell\text{”}} = \sum_{\ell \in S_Y} g(Y) \, e(Y) \, \mathbb{1}_{\text{“}Y=\ell\text{”}}$$

$$= g(Y) \, e(Y) \sum_{\ell \in S_Y} \mathbb{1}_{\text{“}Y=\ell\text{”}}$$

Valore atteso condizionato (caso discreto)

Definizione

$$\mathbb{E}[X \mid Y] = \sum_{k \in S_X} k \, p_{X \mid Y}(k \mid Y)$$

PROPRIETÀ:

- $\mathbb{E}[\mathbb{E}[X \mid Y]] = \mathbb{E}[X]$
- $\mathbb{E}[g(Y) X \mid Y] = g(Y) \mathbb{E}[X \mid Y]$:

$$\mathbb{E}[g(Y) X \mid Y] = \sum_{\ell \in S_Y} \mathbb{E}[g(Y) X \mid Y = \ell] \cdot \mathbb{1}_{\text{“}Y=\ell\text{”}}$$

$$= \sum_{\ell \in S_Y} g(\ell) \underbrace{\mathbb{E}[X \mid Y = \ell]}_{e(\ell)} \cdot \mathbb{1}_{\text{“}Y=\ell\text{”}} = \sum_{\ell \in S_Y} g(Y) \, e(Y) \, \mathbb{1}_{\text{“}Y=\ell\text{”}}$$

$$= g(Y) \, e(Y) \sum_{\ell \in S_Y} \mathbb{1}_{\text{“}Y=\ell\text{”}} = g(Y) \, e(Y) \cdot 1$$

Valore atteso condizionato (caso discreto)

Definizione

$$\mathbb{E}[X \mid Y] = \sum_{k \in S_X} k \, p_{X \mid Y}(k \mid Y)$$

PROPRIETÀ:

- $\mathbb{E}[\mathbb{E}[X \mid Y]] = \mathbb{E}[X]$
- $\mathbb{E}[g(Y) X \mid Y] = g(Y) \mathbb{E}[X \mid Y]$:

$$\mathbb{E}[g(Y) X \mid Y] = \sum_{\ell \in S_Y} \mathbb{E}[g(Y) X \mid Y = \ell] \cdot \mathbb{1}_{\text{“}Y=\ell\text{”}}$$

$$= \sum_{\ell \in S_Y} g(\ell) \underbrace{\mathbb{E}[X \mid Y = \ell]}_{e(\ell)} \cdot \mathbb{1}_{\text{“}Y=\ell\text{”}} = \sum_{\ell \in S_Y} g(Y) \, e(Y) \, \mathbb{1}_{\text{“}Y=\ell\text{”}}$$

$$= g(Y) \, e(Y) \sum_{\ell \in S_Y} \mathbb{1}_{\text{“}Y=\ell\text{”}} = g(Y) \, e(Y) \cdot 1 = g(Y) \, \mathbb{E}[X \mid Y]$$

Valore atteso condizionato (caso discreto)

Definizione

$$\mathbb{E}[X \mid Y] = \sum_{k \in S_X} k \, p_{X \mid Y}(k \mid Y)$$

PROPRIETÀ:

- $\mathbb{E}[\mathbb{E}[X \mid Y]] = \mathbb{E}[X]$
- $\mathbb{E}[g(Y)\,X \mid Y] = g(Y)\,\mathbb{E}[X \mid Y]$:

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$$= \sum_{\ell \in S_Y} g(\ell) \underbrace{\mathbb{E}[X \mid Y = \ell]}_{e(\ell)} \cdot \mathbb{1}_{\text{“}Y=\ell\text{”}} = \sum_{\ell \in S_Y} g(Y)\,e(Y)\,\mathbb{1}_{\text{“}Y=\ell\text{”}}$$

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$$\mathbb{E}\big[\mathbb{E}[X \mid Y] \mid Y\big] = \mathbb{E}\big[e(Y) \mid Y\big] \quad \text{con} \quad e(Y) = \mathbb{E}[X \mid Y]$$

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Definizione

$$\mathbb{E}[X \mid Y] = \sum_{k \in S_X} k \, p_{X \mid Y}(k \mid Y)$$

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$$= e(Y)\, \mathbb{E}\big[1 \mid Y\big]$$

Valore atteso condizionato (caso discreto)

Definizione

$$\mathbb{E}[X \mid Y] = \sum_{k \in S_X} k \, p_{X \mid Y}(k \mid Y)$$

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$$= e(Y)\, \mathbb{E}\big[1 \mid Y\big] = e(Y) \cdot 1$$

Valore atteso condizionato (caso discreto)

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- X e Y indipendenti $\Rightarrow$ $\mathbb{E}[X \mid Y] = \mathbb{E}[X]$ (v.a. costante)

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Non vale il viceversa !

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perché X, Y indip.

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perché X, Y indip.

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- X e Y indipendenti $\Rightarrow$ $\mathbb{E}[X \mid Y] = \mathbb{E}[X]$ (v.a. costante) :

$$\mathbb{E}[X \mid Y] = \sum_{\ell \in S_Y} \mathbb{E}[X \mid Y = \ell] \cdot \mathbb{1}_{\text{“}Y=\ell\text{”}} = \mathbb{E}[X] \underbrace{\sum_{\ell \in S_Y} \mathbb{1}_{\text{“}Y=\ell\text{”}}}_{1} = \mathbb{E}[X]$$

Valore atteso condizionato (caso discreto)

Definizione

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- $\mathbb{E}\Big[\big(X - \mathbb{E}[X \mid Y]\big)^2\Big] \leq \mathbb{E}\Big[\big(X - g(Y)\big)^2\Big]$ per ogni funzione g

  $\big(\, g(Y) = \mathbb{E}[X \mid Y]$ minimizza lo scarto quadratico medio da $X \,\big)$

Valore atteso condizionato (caso discreto)

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  $$\mathbb{E}\Big[\big(\mathbb{E}[X \mid Y] - g(Y)\big)\big(X - \mathbb{E}[X \mid Y]\big)\Big] = \; ???$$

Valore atteso condizionato (caso discreto)

Definizione

$$\mathbb{E}[X \mid Y] = \sum_{k \in S_X} k \, p_{X \mid Y}(k \mid Y)$$

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$$\mathbb{E}\Big[\big(\mathbb{E}[X \mid Y] - g(Y)\big)\big(X - \mathbb{E}[X \mid Y]\big)\Big] =$$

$$= \mathbb{E}\Big[\,\mathbb{E}\Big[\big(\mathbb{E}[X \mid Y] - g(Y)\big)\big(X - \mathbb{E}[X \mid Y]\big) \mid Y\Big]\,\Big]$$

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$$= \mathbb{E}\Big[\,\mathbb{E}\Big[\big(\mathbb{E}[X \mid Y] - g(Y)\big)\big(X - \mathbb{E}[X \mid Y]\big) \mid Y\Big]\,\Big]$$

$$= \mathbb{E}\Big[\,\big(\mathbb{E}[X \mid Y] - g(Y)\big)\,\mathbb{E}\Big[\big(X - \mathbb{E}[X \mid Y]\big) \mid Y\Big]\,\Big]$$

Valore atteso condizionato (caso discreto)

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$$= \mathbb{E}\Big[\, \mathbb{E}\Big[\big(\mathbb{E}[X \mid Y] - g(Y)\big)\big(X - \mathbb{E}[X \mid Y]\big) \mid Y\Big] \,\Big]$$

$$= \mathbb{E}\Big[\, \big(\mathbb{E}[X \mid Y] - g(Y)\big)\, \mathbb{E}\Big[\big(X - \mathbb{E}[X \mid Y]\big) \mid Y\Big] \,\Big]$$

$$= \mathbb{E}\Big[\, \big(\mathbb{E}[X \mid Y] - g(Y)\big)\, \big(\mathbb{E}[X \mid Y] - \mathbb{E}[\,\mathbb{E}[X \mid Y] \mid Y]\big) \,\Big]$$

Valore atteso condizionato (caso discreto)

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$$= \mathbb{E}\Big[\, \big(\mathbb{E}[X \mid Y] - g(Y)\big)\, \mathbb{E}\Big[\big(X - \mathbb{E}[X \mid Y]\big) \mid Y\Big] \,\Big]$$

$$= \mathbb{E}\Big[\, \big(\mathbb{E}[X \mid Y] - g(Y)\big)\, \big(\mathbb{E}[X \mid Y] - \cancel{\mathbb{E}[}\, \mathbb{E}[X \mid Y] \cancel{\mid Y]}\big) \,\Big]$$

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$$= \mathbb{E}\Big[\,\mathbb{E}\Big[\big(\mathbb{E}[X \mid Y] - g(Y)\big)\big(X - \mathbb{E}[X \mid Y]\big) \mid Y\Big]\,\Big]$$

$$= \mathbb{E}\Big[\,\big(\mathbb{E}[X \mid Y] - g(Y)\big)\,\mathbb{E}\Big[\big(X - \mathbb{E}[X \mid Y]\big) \mid Y\Big]\,\Big]$$

$$= \mathbb{E}\Big[\,\big(\mathbb{E}[X \mid Y] - g(Y)\big)\,\big(\mathbb{E}[X \mid Y] - \mathbb{E}[\,\mathbb{E}[X \mid Y] \mid Y]\big)\,\Big] = 0$$

Valore atteso condizionato (caso discreto)

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$$\mathbb{E}\Big[\big(\mathbb{E}[X \mid Y] - g(Y)\big)\big(X - \mathbb{E}[X \mid Y]\big)\Big] =$$

$$= \mathbb{E}\Big[\, \mathbb{E}\Big[\big(\mathbb{E}[X \mid Y] - g(Y)\big)\big(X - \mathbb{E}[X \mid Y]\big) \mid Y\Big] \,\Big]$$

$$= \mathbb{E}\Big[\, \big(\mathbb{E}[X \mid Y] - g(Y)\big)\, \mathbb{E}\Big[\big(X - \mathbb{E}[X \mid Y]\big) \mid Y\Big] \,\Big]$$

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  $\mathbb{E}\Big[\big(X - g(Y)\big)^2\Big] = ???$

Valore atteso condizionato (caso discreto)

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- $\mathbb{E}\Big[\big(X - \mathbb{E}[X \mid Y]\big)^2\Big] \leq \mathbb{E}\Big[\big(X - g(Y)\big)^2\Big]$ per ogni funzione g:

$$\mathbb{E}\Big[\big(X - g(Y)\big)^2\Big] = \mathbb{E}\Big[\big(X - \mathbb{E}[X \mid Y] + \mathbb{E}[X \mid Y] - g(Y)\big)^2\Big]$$

Valore atteso condizionato (caso discreto)

Definizione

$$\mathbb{E}[X \mid Y] = \sum_{k \in S_X} k \, p_{X \mid Y}(k \mid Y)$$

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- $\mathbb{E}\Big[\big(X - \mathbb{E}[X \mid Y]\big)^2\Big] \leq \mathbb{E}\Big[\big(X - g(Y)\big)^2\Big]$ per ogni funzione g:

$$\begin{aligned}
\mathbb{E}\Big[\big(X - g(Y)\big)^2\Big] &= \mathbb{E}\Big[\big(X - \mathbb{E}[X \mid Y] + \mathbb{E}[X \mid Y] - g(Y)\big)^2\Big] \\
&= \mathbb{E}\Big[\big(X - \mathbb{E}[X \mid Y]\big)^2\Big] + \mathbb{E}\Big[\big(\mathbb{E}[X \mid Y] - g(Y)\big)^2\Big] + \\
&\quad + 2\,\mathbb{E}\Big[\big(X - \mathbb{E}[X \mid Y]\big)\big(\mathbb{E}[X \mid Y] - g(Y)\big)\Big]
\end{aligned}$$

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\mathbb{E}\Big[\big(X - g(Y)\big)^2\Big] &= \mathbb{E}\Big[\big(X - \mathbb{E}[X \mid Y] + \mathbb{E}[X \mid Y] - g(Y)\big)^2\Big] \\
&= \mathbb{E}\Big[\big(X - \mathbb{E}[X \mid Y]\big)^2\Big] + \mathbb{E}\Big[\big(\mathbb{E}[X \mid Y] - g(Y)\big)^2\Big] + \\
&\quad + 2\,\mathbb{E}\Big[\big(X - \mathbb{E}[X \mid Y]\big)\big(\mathbb{E}[X \mid Y] - g(Y)\big)\Big] \\
&= \mathbb{E}\Big[\big(X - \mathbb{E}[X \mid Y]\big)^2\Big] + \mathbb{E}\Big[\big(\mathbb{E}[X \mid Y] - g(Y)\big)^2\Big]
\end{aligned}$$

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\mathbb{E}\Big[\big(X - g(Y)\big)^2\Big] &= \mathbb{E}\Big[\big(X - \mathbb{E}[X \mid Y] + \mathbb{E}[X \mid Y] - g(Y)\big)^2\Big] \\
&= \mathbb{E}\Big[\big(X - \mathbb{E}[X \mid Y]\big)^2\Big] + \mathbb{E}\Big[\big(\mathbb{E}[X \mid Y] - g(Y)\big)^2\Big] + \\
&\quad + 2\,\cancel{\mathbb{E}\Big[\big(X - \mathbb{E}[X \mid Y]\big)\big(\mathbb{E}[X \mid Y] - g(Y)\big)\Big]} \\
&= \mathbb{E}\Big[\big(X - \mathbb{E}[X \mid Y]\big)^2\Big] + \mathbb{E}\Big[\big(\mathbb{E}[X \mid Y] - g(Y)\big)^2\Big] \\
&\geq \mathbb{E}\Big[\big(X - \mathbb{E}[X \mid Y]\big)^2\Big]
\end{aligned}$$

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$$\begin{aligned}
\mathbb{E}\Big[\big(X - g(Y)\big)^2\Big] &= \mathbb{E}\Big[\big(X - \mathbb{E}[X \mid Y] + \mathbb{E}[X \mid Y] - g(Y)\big)^2\Big] \\
&= \mathbb{E}\Big[\big(X - \mathbb{E}[X \mid Y]\big)^2\Big] + \mathbb{E}\Big[\big(\mathbb{E}[X \mid Y] - g(Y)\big)^2\Big] + \\
&\quad + 2\,\cancel{\mathbb{E}\Big[\big(X - \mathbb{E}[X \mid Y]\big)\big(\mathbb{E}[X \mid Y] - g(Y)\big)\Big]} \\
&= \mathbb{E}\Big[\big(X - \mathbb{E}[X \mid Y]\big)^2\Big] + \mathbb{E}\Big[\big(\mathbb{E}[X \mid Y] - g(Y)\big)^2\Big] \\
&\geq \mathbb{E}\Big[\big(X - \mathbb{E}[X \mid Y]\big)^2\Big]
\end{aligned}$$

Valore atteso condizionato (caso discreto)

Definizione

$$\mathbb{E}[X \mid Y] = \sum_{k \in S_X} k \, p_{X \mid Y}(k \mid Y)$$

PROPRIETÀ:

- $\mathbb{E}\Big[\big(X - \mathbb{E}[X \mid Y]\big)^2\Big] \leq \mathbb{E}\Big[\big(X - g(Y)\big)^2\Big]$ per ogni funzione g

Densità condizionate (caso assolutamente continuo)

Definizione

(i) $(\vec{X}, Y)$: vettore aleatorio ass. continuo $(n+1)$-dimensionale

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PROPRIETÀ: simili al caso discreto, con $\sum \to \int$

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